

# DATA ANALYSIS

## Milestone Project TATA POWER

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PANKAJ SINGH

## Page 1: Business and Usage Overview

Problems to address:

- How is charging demand evolving over time
- Are stations able to handle growing demand
- What is the charging behaviour pattern by time of day

### KPIs to track:

- Total charging sessions
  - Total energy consumed (kWh charged)
  - Total revenue (from charging sessions)
  - Number of active customers
  - Average revenue per session
  - Average kWh per session
-

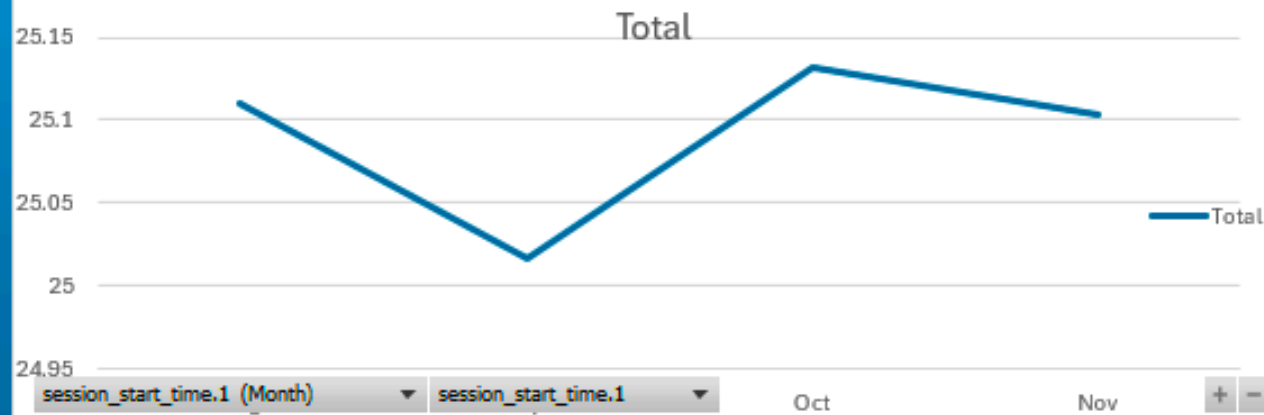
## Business and Usage Overview

TATA POWER  
PROJECT114000  
Total charging sessions8000  
Number of active customers2859771.70  
Total energy consumed (kWh charged)435.91  
Average revenue per session49693947.04  
Total revenue (from charging sessions)25.09  
Average kWh per session

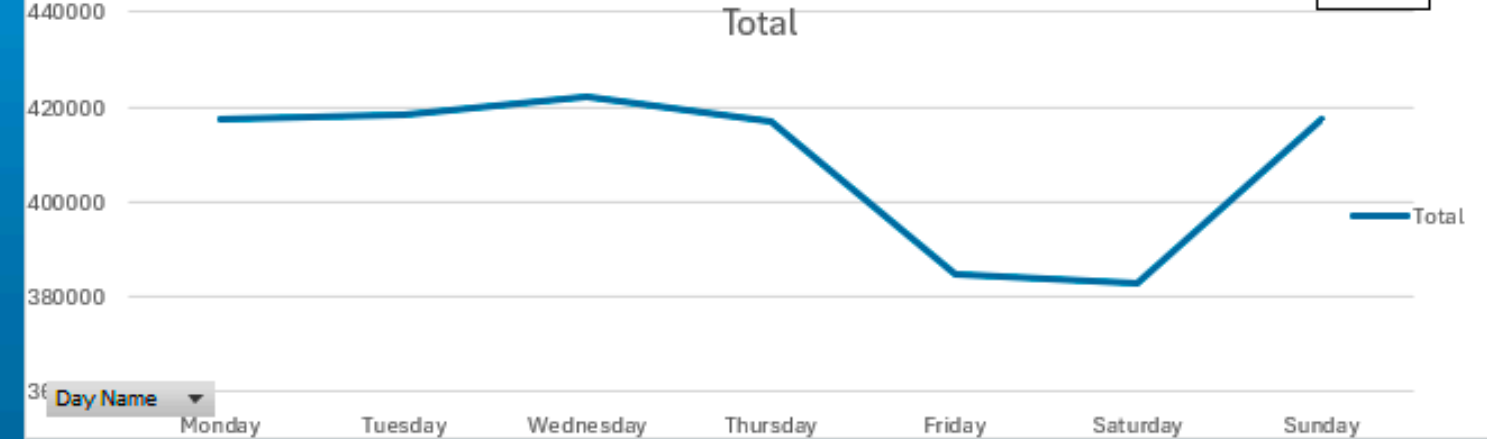
Day Name

Friday  
Monday  
Saturday  
Sunday  
Thursday  
Tuesday  
Wednesday

Average of kwh\_charged



Sum of kwh\_charged

Total  
revenue

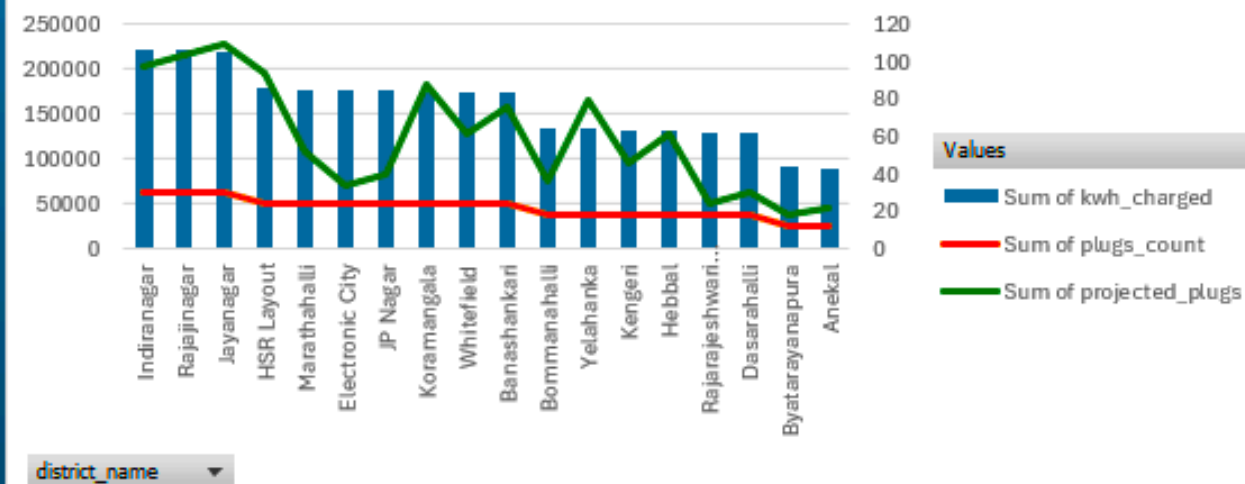
district\_name

JP Nagar  
Kengeri  
Koramangala  
Marathahalli  
Rajajinagar  
Rajarajeshwari Na...  
Whitefield  
Yelahanka

Sum of kwh\_charged

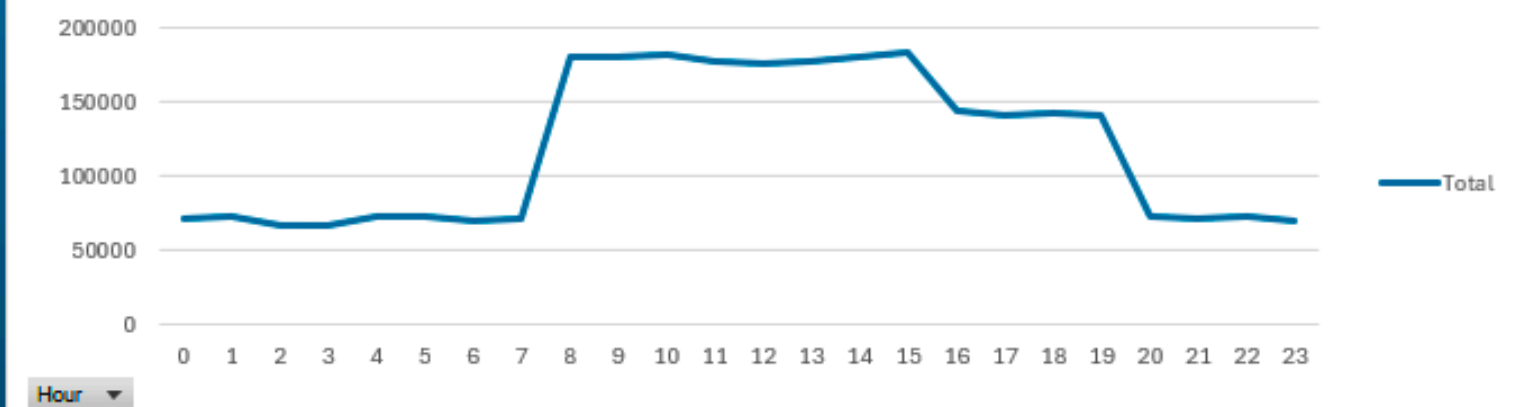
Sum of plugs\_count

Sum of projected\_plugs



Sum of kwh\_charged

Total



This dashboard analyzes EV charging usage patterns, station utilization, and customer charging behavior to understand operational efficiency and future infrastructure needs.

1. Charging Demand Evolution Over Time  
Explanation:

The dashboard shows that EV charging demand remains consistently high across the observed period. Total charging sessions reached 114,000, with more than 2.85 million kWh of energy consumed.

The monthly trend indicates stable charging activity with slight fluctuations, suggesting a steady adoption of EV charging services.

Key Insight:  
Charging demand is stable and continuously active.  
Weekday charging demand is comparatively higher than weekends.  
The business is generating strong usage and revenue growth.

2. Station Capacity and Demand Handling  
Explanation:

District-wise analysis compares energy consumption with available charging plugs and projected plug requirements.

Some districts such as Indiranagar, JP Nagar, and Koramangala show significantly higher charging activity, indicating higher station utilization.

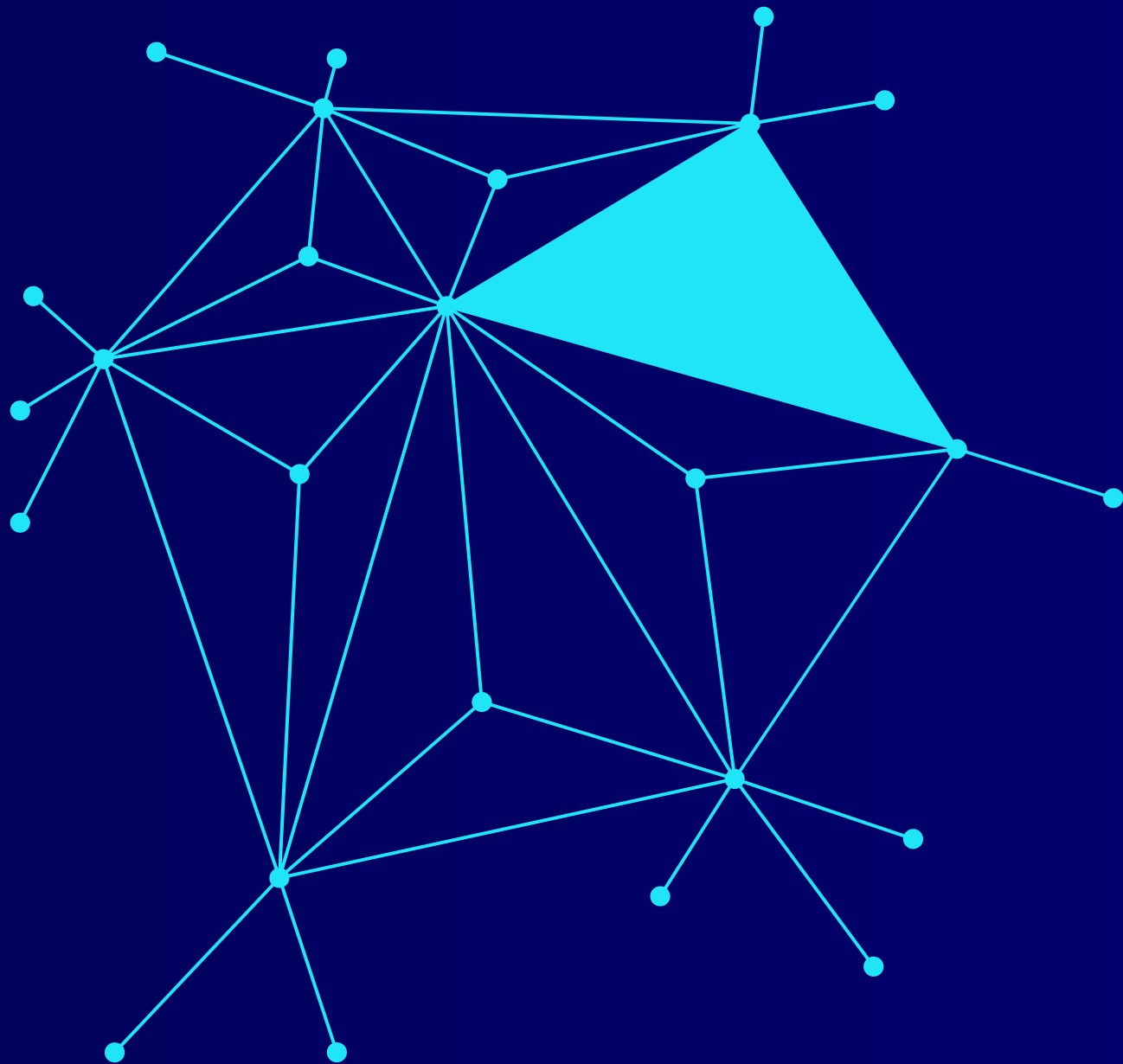
Key Insight:  
High-demand districts may require additional charging infrastructure in the future.  
Certain stations appear underutilized and may need operational optimization.  
Existing infrastructure is handling demand, but future scaling will be important.

3. Charging Behaviour by Time of Day  
Explanation:

The hourly usage trend shows that charging demand peaks during daytime hours, especially between 8 AM and 4 PM.

Charging activity drops significantly during late evening and early morning hours.

Key Insight:  
Most users charge vehicles during working/business hours.  
Charging behavior suggests strong commuter-based usage patterns.  
Peak-hour congestion may occur during daytime periods.  
Overall Business Insights  
Total Revenue generated: 49.69 Million  
Average Revenue per Session: 435.91  
Average Energy Consumption per Session: 25.09 kWh  
Active Customer Base: 8,000 users



## **Page 2: Station Utilization and Operational Efficiency**

### **Problems to address:**

- Which stations are overutilized and may require capacity expansion
- Which stations are underutilized and need operational improvement
- Are available plugs being efficiently used across locations
- What is the load distribution across districts and stations

### **KPIs to track:**

- Number of sessions per station
  - Average utilization per station (sessions per plug)
  - Plugs count vs actual usage
  - Average kWh per station
  - District wise station density
  - Sessions per plug ratio
-



TATA POWER  
PROJECT

1754  
number of session per station

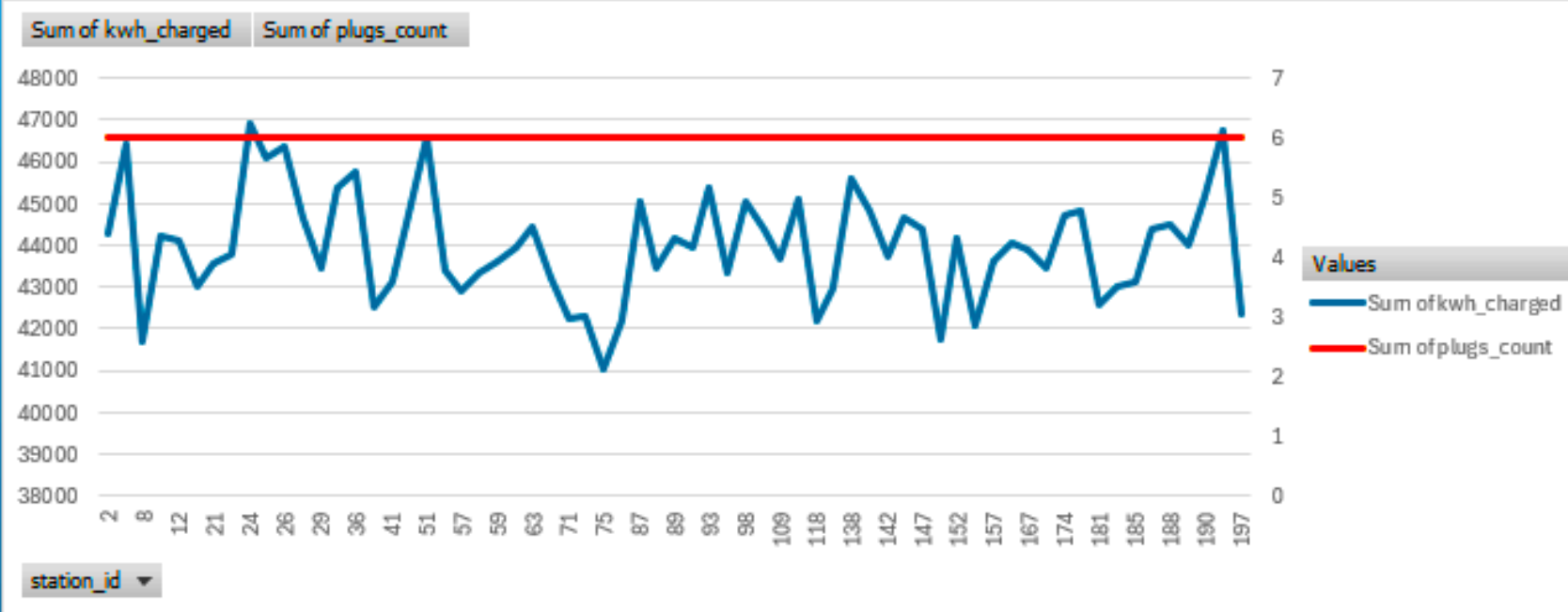
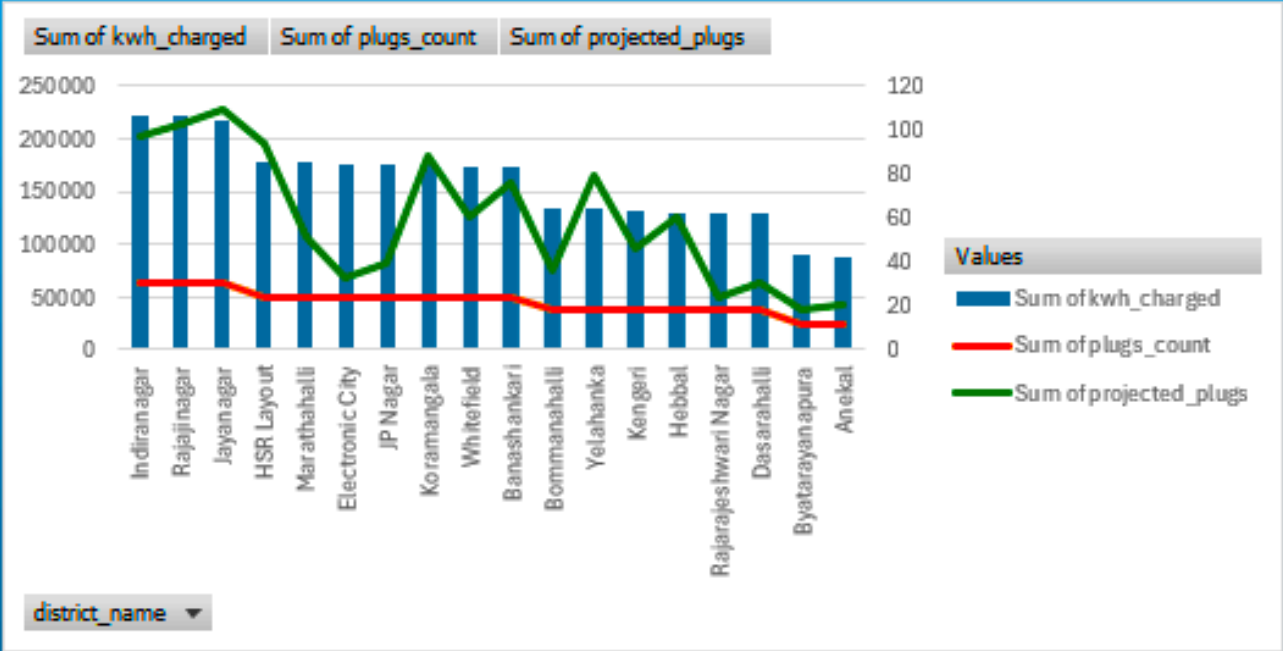
1753.85  
average utilization per station (session per plug)

7332.75  
plug count vs actual usage

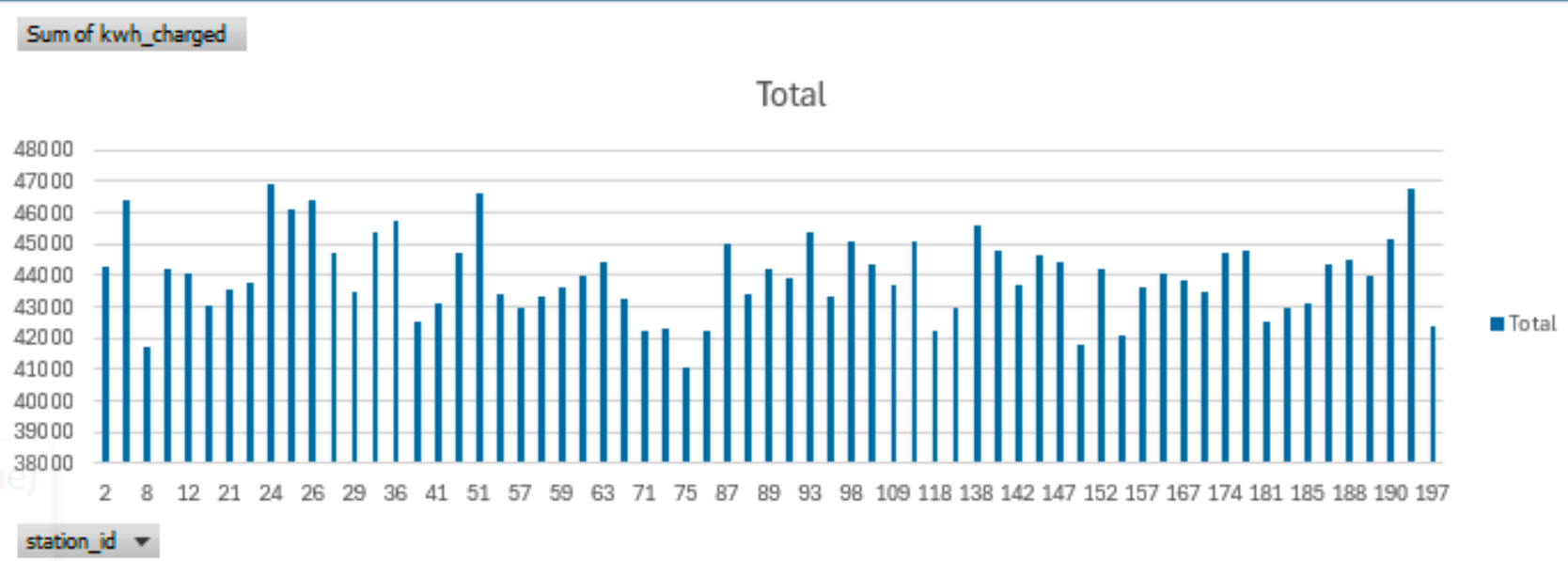
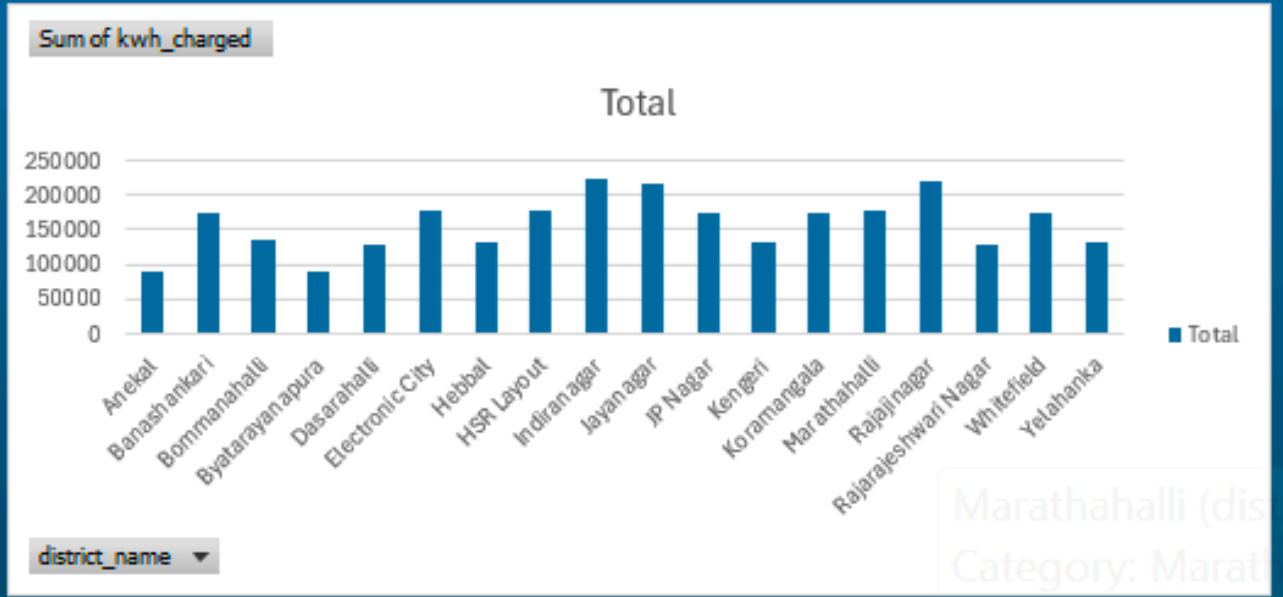
25.09  
Average KWH per station

3.61  
district wise station

- district\_name
- Anekal
  - Banashankari
  - Bommanahalli
  - Byatarayanapura
  - Dasarahalli
  - Electronic City
  - Hebbal
  - HSR Layout
  - Indiranagar



- district\_name
- JP Nagar
  - Kengeri
  - Koramangala
  - Marathahalli
  - Rajajinagar
  - Rajarajeshwari Nagar
  - Whitefield
  - Yelahanka



This dashboard evaluates how efficiently charging stations and plugs are being utilized across different districts and stations. It helps identify overloaded stations, underutilized locations, and future infrastructure requirements.

1. Station Utilization Analysis

Explanation:

The dashboard tracks station-level charging activity using metrics such as sessions per station, average utilization, and energy consumption.

The analysis shows variation in charging demand across locations, with some stations consistently handling higher charging loads than others.

Key Insight:

High-demand stations are operating at greater utilization levels.  
Certain stations may require future capacity expansion to handle increasing EV demand.  
Utilization patterns vary significantly across districts.

2. Overutilized vs Underutilized Stations

Explanation:

The station-wise utilization charts compare actual charging usage with available plug capacity.

Stations with high charging activity and limited plug availability indicate possible overutilization, while stations with lower charging demand may be underutilized.

Key Insight:

Stations in major commercial and residential areas show higher usage intensity.  
Underutilized stations may require better operational planning or customer awareness.  
Capacity balancing across stations can improve operational efficiency.

3. Plug Efficiency Across Locations

Explanation:

The dashboard compares total plugs available against actual charging usage to evaluate whether charging infrastructure is being efficiently used.

Average utilization per station is approximately 1753.85 sessions per plug, indicating active infrastructure usage.

Key Insight:

Plug utilization remains strong across most high-demand locations.  
Some locations may require additional plugs to avoid congestion during peak demand.  
Efficient plug distribution is critical for maintaining service quality.

4. Load Distribution Across Districts and Stations

Explanation:

District-wise and station-wise energy consumption analysis highlights how charging demand is distributed geographically.

Districts such as Indiranagar, JP Nagar, and Rajajinagar show comparatively higher charging loads, indicating strong EV activity in these areas.

Key Insight:

Charging demand is concentrated in urban and commercial districts.  
Balanced infrastructure expansion should focus on high-load regions.  
Certain districts have lower station density despite increasing usage trends.

KPI Highlights

Number of Sessions per Station: 1754

Average Utilization per Station: 1753.85

Average kWh per Station: 25.09

Plug Count vs Actual Usage: 7332.75

District-wise Station Density: 3.61

Final Conclusion

The dashboard indicates that charging infrastructure utilization is uneven across locations. High-demand districts may require additional charging capacity, while underperforming stations need operational optimization. Improving plug allocation and balancing station loads will help support future EV charging growth efficiently.

## Page 3: Customer and Demand Behavior

### Problems to address:

- What type of customers contribute most to usage and revenue
- How does income tier influence charging behavior
- What are the usage patterns across different EV models and battery capacities
- Are there segments with high potential but low engagement

### KPIs to track:

- Unique customers count
  - Sessions per customer
  - Average kWh per customer
  - Revenue per customer
-



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PROJECT

8000  
Unique customer count

14  
session per customer

357  
average kwh per customer

6212  
revenue per customer

income\_tier



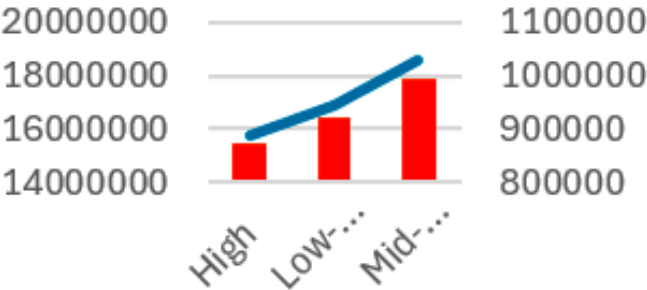
High

Low-Mid

Mid-Range

Sum of kwh\_charged

Sum of total\_cost



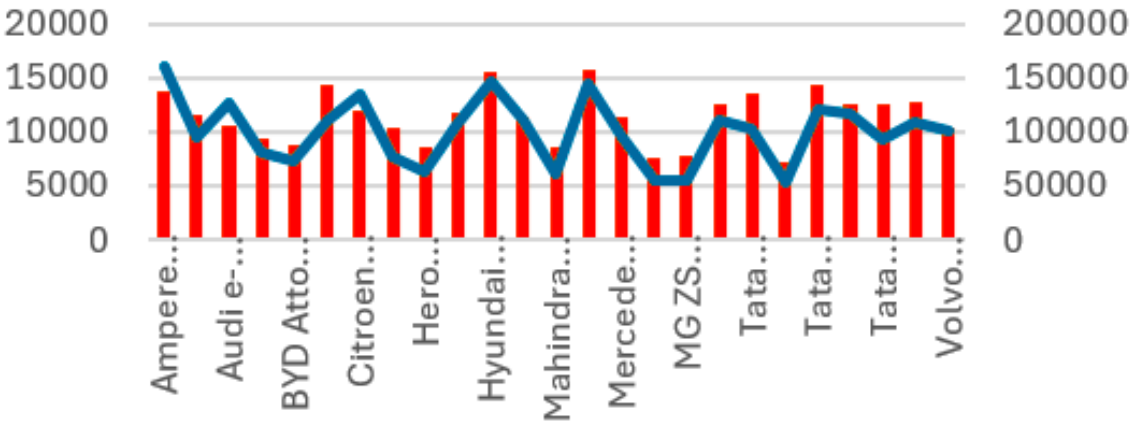
Values

Sum of...

income\_tier

Sum of battery\_capacity\_kwh

Sum of kwh\_charged



car\_model

Sum of kwh\_charged

Total

Mid-Range  
36%

High  
31%

Low-Mid  
33%

income\_tier

High

Low-Mid

Mid-Range

Page 3: Customer and Demand Behavior — PPT Explanation  
Objective

This dashboard analyzes customer charging behavior, revenue contribution, and EV usage patterns across different income groups and vehicle models.

1. Customer Contribution to Usage and Revenue  
Explanation:

The dashboard evaluates how different customer segments contribute to charging activity and revenue generation.

A total of 8,000 unique customers used the charging network. On average:

Each customer performed 14 charging sessions  
Average energy consumption per customer was 357 kWh  
Average revenue generated per customer was 6212  
Key Insight:  
Customers show strong repeat charging behavior.  
The charging network is generating stable customer usage and revenue.  
Revenue contribution is distributed across multiple customer groups.  
2. Impact of Income Tier on Charging Behavior  
Explanation:

Charging demand and charging cost were analyzed across High, Low-Mid, and Mid-Range income groups.

The analysis indicates that Mid-Range and Low-Mid income groups contribute a significant share of total charging activity.

Key Insight:  
EV charging adoption is spread across different income segments.  
Mid-Range customers contribute the highest charging demand.  
Charging behavior remains relatively balanced across income categories.  
3. Usage Distribution by Income Tier  
Explanation:

The pie chart represents the percentage contribution of each income tier to overall charging usage.

Mid-Range customers contribute approximately 36%  
Low-Mid customers contribute around 33%  
High-income customers contribute nearly 31%  
Key Insight:  
Mid-Range customers are the largest contributors to charging usage.  
The customer base is diversified across all income groups.  
No single income segment dominates the charging network.  
4. EV Model and Battery Capacity Usage Pattern  
Explanation:

The dashboard compares battery capacity and charging consumption across different EV models.

Vehicles with larger battery capacities generally show higher charging energy consumption.

Key Insight:  
EV models with higher battery capacity consume more charging energy.  
Certain vehicle models contribute significantly to network utilization.  
Vehicle-level demand analysis can support future charging infrastructure planning.  
KPI Highlights  
Unique Customers: 8,000  
Sessions per Customer: 14  
Average kWh per Customer: 357  
Revenue per Customer: 6212  
Final Conclusion

The dashboard highlights strong customer engagement and balanced charging behavior across different income groups. Mid-Range customers contribute the highest charging demand, while EV model analysis helps identify vehicle segments driving overall energy consumption and charging network usage.

## Page 4: Financial Performance and Expansion Strategy (ROI Focus)

Page 11

Problems to address:

- Which districts and projects deliver the highest ROI
- What is the payback period for different charging setups
- How do costs such as CAPEX, land lease, and infrastructure impact profitability

### KPIs to track:

- Total CAPEX per project
- Annual revenue and EBITDA
- Payback period (with and without subsidy)
- 5 year ROI percentage
- Existing stations vs projected EVs
- Demand supply gap (projected EVs vs projected plugs)





TATA POWER  
PROJECT

142  
Total CAPEX per project

4142.86%  
5 year ROI percentage

4629, 3239  
Annual revenue and EBITDA

610  
Existing station VS project  
EVs

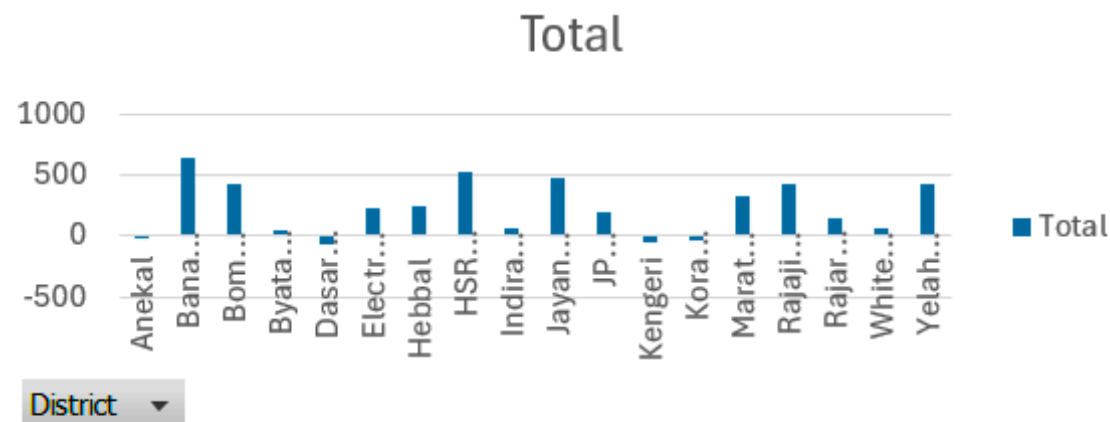
451.3  
Payback period (with and  
without subsidy)

33  
demand supply gap

district...

Anekal  
Banashan...  
Bomman...  
Byataraya...  
Dasarahalli

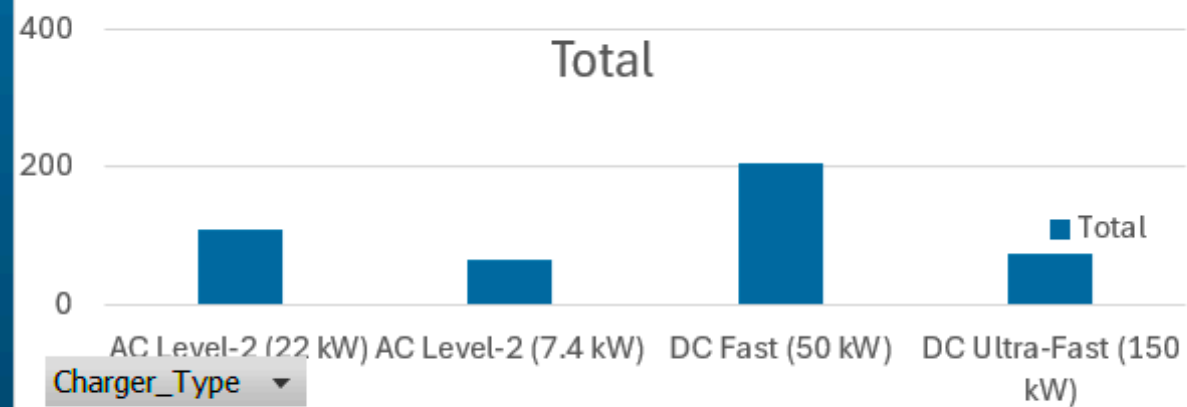
Sum of 5yr\_ROI\_Pct



district\_name

Anekal  
Banashankari  
Bommanahalli  
Byatarayanapura  
Dasarahalli  
Electronic City  
Hebbal  
HSR Layout  
Indiranagar  
Jayanagar  
JP Nagar  
Kengeri

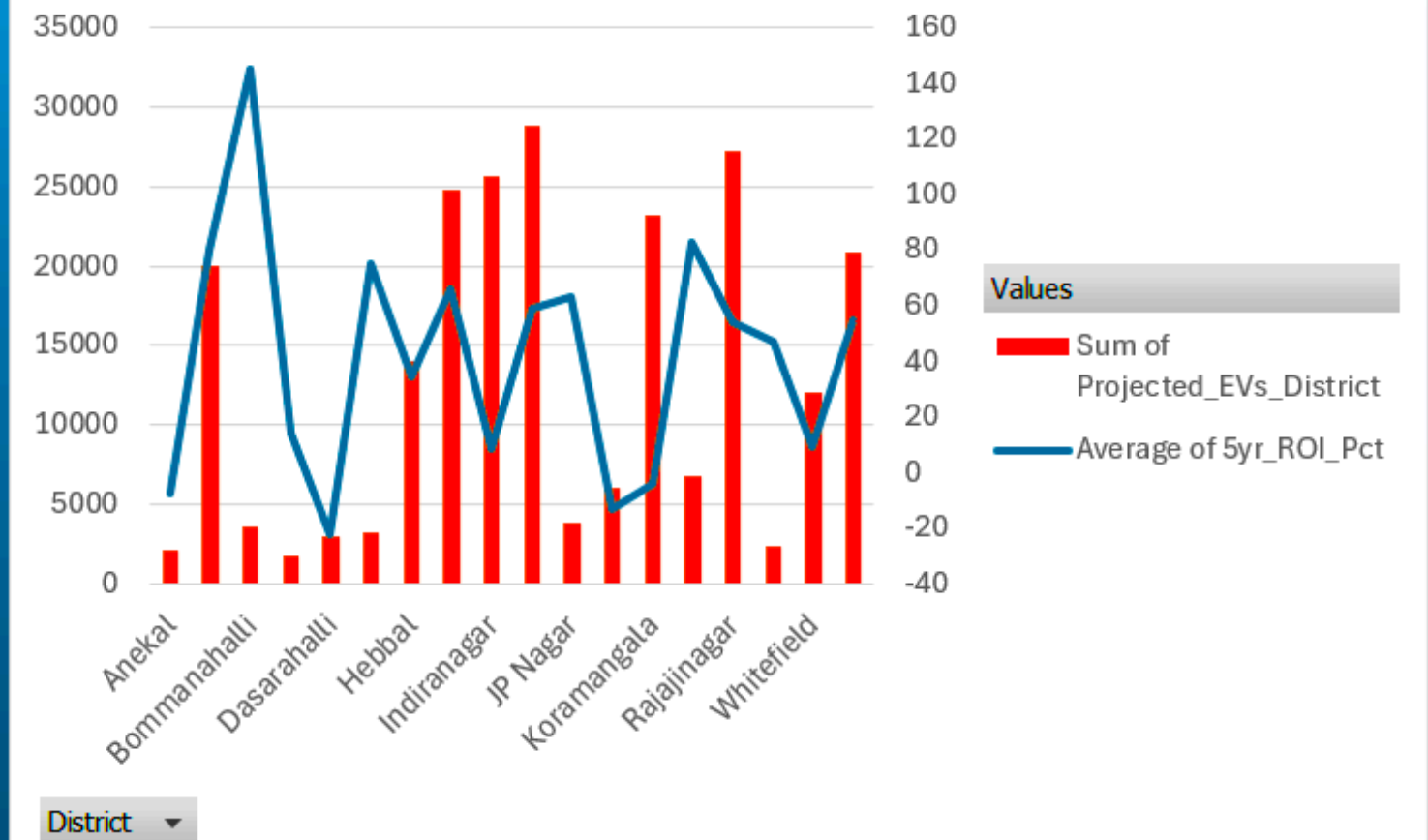
Sum of Payback\_Period\_Yrs



Charger\_Type

AC Level-2 (22 kW)  
AC Level-2 (7.4 kW)  
DC Fast (50 kW)  
DC Ultra-Fast (150...)

Average of 5yr\_ROI\_Pct Sum of Projected\_EVs\_District



- Future Scope / Expansion Strategy (ROI Focus)
- Future Plans & Strategic Actions
- Tata Power will focus on expanding EV charging infrastructure in districts with the highest ROI and fastest EV adoption growth.
- DC Fast and Ultra-Fast chargers will be prioritized in high-demand locations to maximize revenue generation and improve charging efficiency.
- Government subsidies, smart investment planning, and public-private partnership (PPP) models will be utilized to reduce the overall payback period.
- Underperforming stations will be optimized to improve utilization rates, operational efficiency, and overall profitability.
- New charging hubs will be developed in districts with high demand-supply gaps to enhance customer accessibility and service availability.
- Strategic partnerships with malls, metro stations, highways, IT parks, and commercial hubs will support large-scale network expansion.
- Solar energy integration and battery storage systems will be implemented to reduce electricity costs and improve EBITDA margins.
- AI-based smart monitoring and predictive maintenance systems will help reduce downtime and maintenance expenses.
- Expansion decisions will be fully data-driven, based on ROI trends, EV growth, charger utilization, revenue performance, and market demand.
- The long-term vision is to build a scalable, sustainable, and highly profitable EV charging ecosystem capable of supporting future EV growth.

Milestone Project

# THANK YOU

FOR YOUR ATTENTION

